

Supplementary Material for HR²-KILO

I. Dataset Configuration

To make the robot motion not limited to a range of the lab, we performed real-world tests, including multiple scenes and different gaits. The details of each sequence used in our experiments are shown in Table S1.

Table S1: DATASET CONFIGURATION

Dataset (Robot)	Sequence ID	Duration (min:sec)	LiDAR DEG ¹	Return Origin	Max Vel (m/sec)	Validated Module ²	Scene/Gait Characteristics
LIKO (BHR-B3)	<i>forward backward</i>	1 : 25	–	✓	–	CD/Odo	walk
	<i>square walk</i>	2 : 12	–	✓	–	CD/Odo	walk
	<i>square walk long</i>	2 : 54	–	✓	–	CD/Odo	walk
	<i>walk in place</i>	3 : 19	–	✓	–	CD/Odo	walk
	<i>up slope</i>	1 : 59	–	✓	–	CD/Odo	up slope/walk
HR ² -KILO (Unitree G1)	<i>walk & back</i>	0 : 40	–	✓	1.2	CD	walk
	<i>forward running</i>	0 : 22	✓	–	2.0	CD	run
	<i>soft terrain</i>	0 : 29	–	–	1.2	CD	soft terrain/walk
	<i>long corridor</i>	18 : 05	✓	✓	1.2	Odo	long term/walk
	<i>short corridor</i>	5 : 15	✓	✓	1.2	Odo	long term/walk
	<i>switchback running</i>	2 : 30	✓	✓	2.0	Odo	run
	<i>spinning motion</i>	0 : 58	–	✓	1.2	Odo	spinning/walk

¹LiDAR suffers degradation in corridors or under severe ground impacts.

²CD represents sequences for contact detection evaluation, and Odo for odometry evaluation.

II. Experiment Details & Analysis

A. Contact Detection

In this section, we provide more detailed experiments of contact detection. On sequences *walk & back*, *forward running* and *soft terrain*, the ground truth of the real-world tests is manually labeled according to the foot height and IMU fluctuation frame-by-frame. As DCE is a learning-based contact detection method, we additionally collected five sequences and labeled the ground truth for training the model, including *stationary stance*, *step in place*, *walk forward & back*, *walk forward & turn to return*, and *running*. Regarding the GRF method, a moving average filter (100 frames size for 1000 Hz data) is applied to detect the contact stably due to the significant noise. The complete detection of contact events after initialization is used as the evaluation metric and the results are shown in Fig. S1.

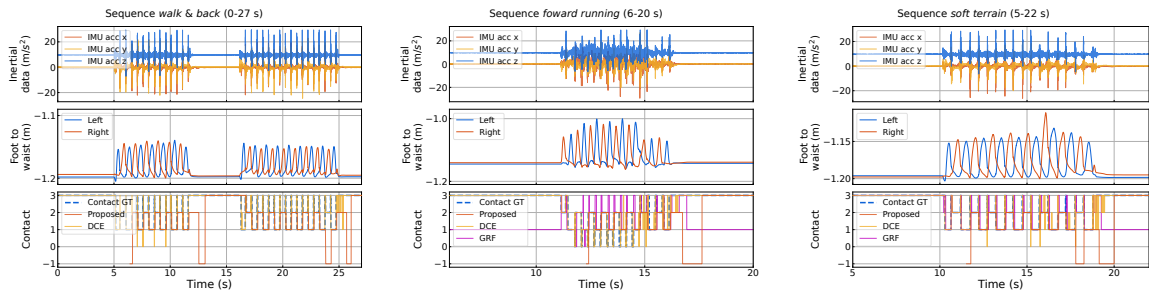


Fig. S1: Comparison experiments on contact detection in the real-world test.

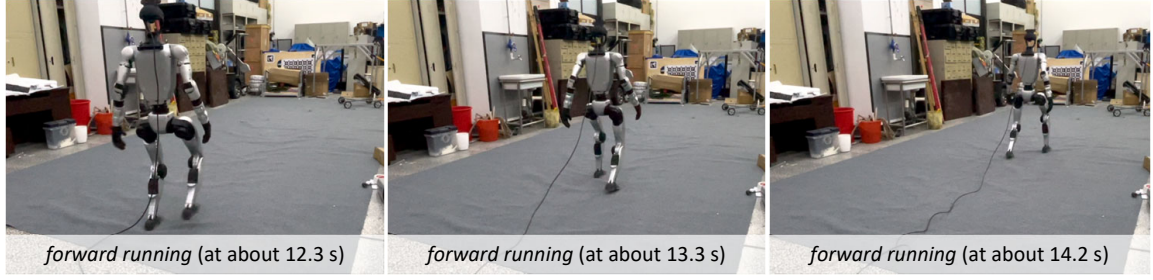


Fig. S2: On *forward running*, the robot runs forward, gradually accelerating at the start and urgently decelerating near the end.



Fig. S3: On *soft terrain*, the robot starts from the marble floor, moves through the carpet, and finally stops on a foam tile.

On the *walk & back* sequence, both the proposed method and DCE achieve stable performance, but the latter shows overdetection after 25 s. Notice that the proposed method is more delayed in detection from single to bipedal contact (at about 13 s and 16 s) to avoid overdetection, which can be set by parameters. The reason that this situation does not degrade the estimator is that using either contact foot as the measurement during the bipedal stance phase will introduce a positive effect.

On *forward running* and *soft terrain* sequences, the proposed method also shows a more stable performance. DCE shows performance degradation after half of both sequences. The reason would be that the dataset does not contain the scenarios during training the model: the robot suffered an emergency deceleration near the end of running (see Fig. S2), and there are no sequences containing soft terrain (similar to Fig. S3) for training. The GRF method with a moving average filter provides stable detection, but introduces additional delay and requires body state and elaborate thresholding.

We also report the FPR and accuracy of contact detection within the state estimation context, as shown in Table S2. Although DCE achieves above 1% lower FPR than ours, its degradation on *forward running* and *soft terrain* reflects a dependence on model training and dataset. In terms of accuracy, the proposed contact module is close to DCE, while GRF performs less competitively. DCE shows superior FPR and accuracy metrics, but a concern is that for robots without contact sensors, obtaining the contact ground truth is difficult and a considerable workload. Given that our method is more general, making the odometry easier to deploy, also with low FPR and favorable accuracy, it remains highly competitive.

There are two limitations of the proposed contact detection module: the slight delay and the inability to sense the dual-foot aerial phases, as shown on sequence *forward running* (Fig. S4). We utilize acceleration fluctuations within a sliding window to trigger contact detection, this process improves detection stability at the cost of a marginal degradation in response capability. Due to the correctly detected contact time significantly exceeding the delay and the contribution

Table S2: FALSE POSITIVE RATE AND ACCURACY OF CONTACT DETECTION

Seq. ID	False Positive Rate (%)			Accuracy (%)		
	Proposed	DCE	GRF	Proposed	DCE	GRF
<i>forward backward</i>	3.09	—	—	94.90	—	—
<i>square walk</i>	4.49	—	—	93.25	—	—
<i>square walk long</i>	4.61	—	—	89.14	—	—
<i>walk in place</i>	3.62	—	—	96.25	—	—
<i>up slope</i>	2.90	—	—	97.65	—	—
<i>walk & back</i>	2.80	1.08	—	93.26	98.81	—
<i>forward running</i>	3.92	3.51	4.72	86.43	94.13	57.62
<i>soft terrain</i>	1.31	2.07	3.81	89.65	93.85	79.73
Average	3.34	2.22	4.27	92.57	95.60	68.68

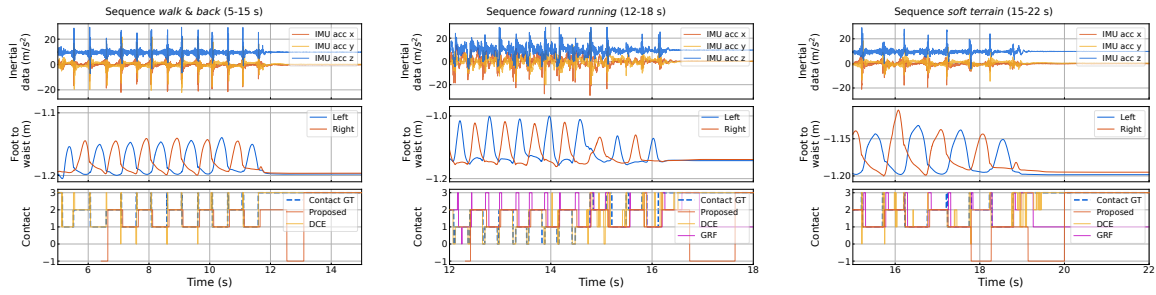


Fig. S4: Details of the comparison experiments on contact detection in the real-world test.

of multi-sensor fusion, it does not degrade the odometry performance. For the latter, the bipedal aerial phase briefly occurs during humanoid robot running and jumping, while it does not constitute the dominant gait. In this work, the general application scenarios of humanoid robots are prioritized.

B. Efficiency Comparison

In this section, we maintain the same parameters to perform the efficiency experiments on the public dataset. We maintain the same parameters to perform the efficiency experiments. The average processing time during each LiDAR scan is computed in Table S3. The proposed method does not have an advantage in comparison. However, the advantages of the high rate and robustness are significant, and the computational efficiency of 25 ms on average is still advanced in fusion odometry.

Table S3: AVERAGE PROCESSING TIME PER SCAN

Methods	<i>forward backward</i>	<i>square walk</i>	<i>square walk long</i>	<i>walk in place</i>	<i>up slope</i>	Average
LIO-SAM	42.314	44.424	42.652	42.643	42.309	42.868
FAST-LIO2	16.241	15.371	15.728	15.124	12.181	14.929
Point-LIO	24.976	26.172	25.391	26.558	17.841	24.188
Point-LIO (H)	29.274	30.904	29.170	30.650	19.793	27.958
LIKO	17.692	17.730	16.851	17.995	12.637	16.581
Proposed	26.163	27.290	26.295	26.878	19.749	25.275
Proposed (H)	29.563	30.944	29.699	30.351	21.900	28.491